

10. The Mirror Cost of English: Exact Palindromic Generation as Bidirectional Re-segmentation

OUTPUT *Abstract.* An exact character-level palindrome requires one letter sequence to support two readings: the usual reading and a reading obtained by reversing the letters and recovering word boundaries. We introduce a symmetric measurement of this burden: re-segment the forward and reversed letters with the same vocabulary and algorithm, then compare their causal-language-model costs per letter. On 57 retained WikiText-2 configurations, the reverse reading costs 2.16–3.60 additional bits per letter; forward dictionary-word coverage is 88–92%, versus 48–55% in reverse. An exact two-ended decoder is a case study: a sound POS-shape gate raises structurally admitted output per popped state by 5.47 times in 50 fixed-work trials, but establishes neither grammaticality nor meaning. The contribution is a protocol for measuring reversal’s linguistic burden, not a claim of coherent long-form generation.

CRITIQUE This final version is appropriately narrow but still pilot-scale: it lacks frozen sampled spans and model revisions, another corpus, an independent model family, and any blinded human study. It should not be submitted as a prose-quality paper.

REVISION APPLIED Keep these limitations in the final manuscript and make the next empirical milestone a preregistered cross-corpus, cross-architecture replication; add human evaluation only if a later paper claims generated-text quality.